

1.

A. Define software interrupt with example.

Interrupt generated by executing an instruction is called software interrupt. It's also called trap.

examples:

1. A cout (or) cin statement would generate a software interrupt because it would make a system call to print something.
2. A fork() statement in Linux would generate a software interrupt because it would make a system call to create a new process.

B. what are the problems of linear decoding system in main memory array design? How to solve them?

Problems of Linear decoding system in main memory array design:

1. **Bus Conflict:** two or more of lines A10-A13 are low at the same time, more than one SRAM chip are selected.

Solution: software must be written such that it never reads into or writes from any address in which more than one of bits A10-A13 are low.

2. **Memory Foldback:** Wastes a large amount of address space. For example, whenever the address value is B800 or 3800, SRAM chip I is selected.

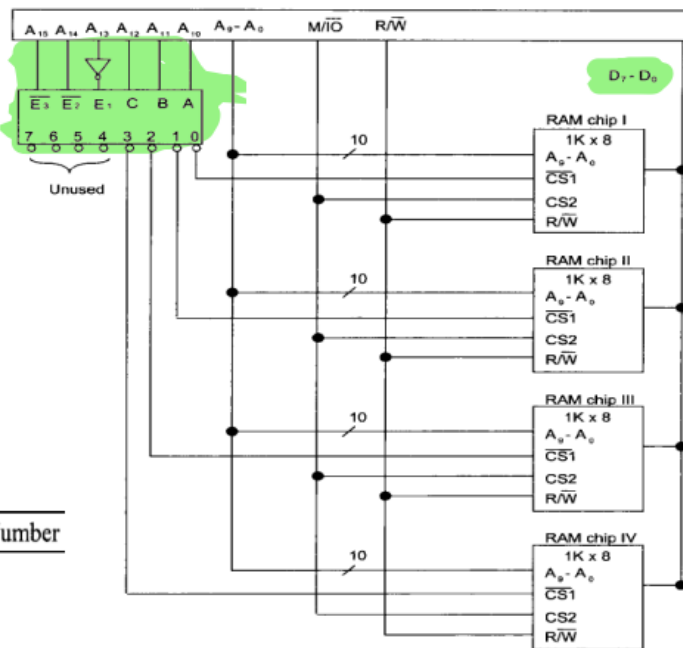
Solution: use full decoded memory addressing.

Main Memory Array Design

- **Full Decoding.**

- Using 3x8 decoder output selects one of the four 1K SRAM chips, depending on the values of A_{12} , A_{11} , and A_{10}
- Decoder output enabled only when $E3 = E2 = 0$ and $E1 = 1$

A_{12}	A_{11}	A_{10}	SRAM Chip Number
0	0	0	I
0	0	1	II
0	1	0	III
0	1	1	IV



Interconnecting a microprocessor with a 4K RAM using full decoded memory addressing.

C. Interfaces two chips of 4k RAM and 1 chip of 2k ROM with processor. Starting address of RAM is 8000H.

A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	
19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
8	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	F	F	F	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1
9	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
9	F	F	F	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
A	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
A	F	F	F	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1
A	8	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0
A	7	F	F	1	0	1	0	0	1	1	1	1	1	1	1	1	1	1	1

Draw the block diagram from the above explanation.

2. A processor having 16 bits address line and it is byte addressable. Interface 4096x8 RAM with the processor.

1. What is the memory capacity of the processor?

The number of address line is 16 bits.

Hence, the memory capacity of the processor = $2^{16}=64k$

2. How many addresses are required for a RAM?

Addresses are required for a RAM = $2^{12}=4k$

3. How many RAM chips are required?

The RAM chips required = $2^{16}/2^{12}=2^4=16$

4. What will be the size of address decoder?

The size of the address decoder is 4 to 16.

Session 2017-2018.

5. B. Differentiate between programmed I/O and interrupt initiated I/O.

Aspect	Programmed I/O	Interrupt-Initiated I/O
Initiation	The CPU directly controls data transfer.	Data transfer is initiated by external hardware signals.
CPU Involvement	CPU actively polls or checks status to transfer data.	CPU is freed from continuously checking; waits for interrupt to handle data transfer.
Efficiency	Less efficient, as CPU needs to continuously check status, tying up resources.	More efficient, as CPU is freed for other tasks until interrupted.
Module	Its module is treated as a slow module.	Its module is faster than programmed I/O module.
Example	A simple example is reading data from a keyboard in which the CPU continuously checks if a key is pressed.	An example is reading data from a network card. The CPU can perform other tasks until the network card signals that data is ready to be read, triggering an interrupt.

6. B. When is an overflow flag set or reset? Give examples.

An overflow flag is set or reset based on the outcome of arithmetic operations, particularly addition and subtraction. It indicates whether the result of an operation fits within the available storage space.

Set: The overflow flag is set when the result of an operation exceeds the maximum value that can be represented in the available storage space.

Example of Overflow Flag Set: In an 8-bit system, if we add two numbers: $200 + 100$, the result is 300. However, since 300 cannot be represented using 8 bits (it's larger than 255), the overflow flag would be set to indicate that the result is too large to fit in the available space.

Reset: The overflow flag is reset when the result of an operation fits within the available storage space.

Example of Overflow Flag Reset: In the same 8-bit system, if you add two numbers: $100 + 50$, the result is 150. Since 150 can be represented using 8 bits, the overflow flag would remain reset because the result fits within the available storage space.

Session 2016-2017.

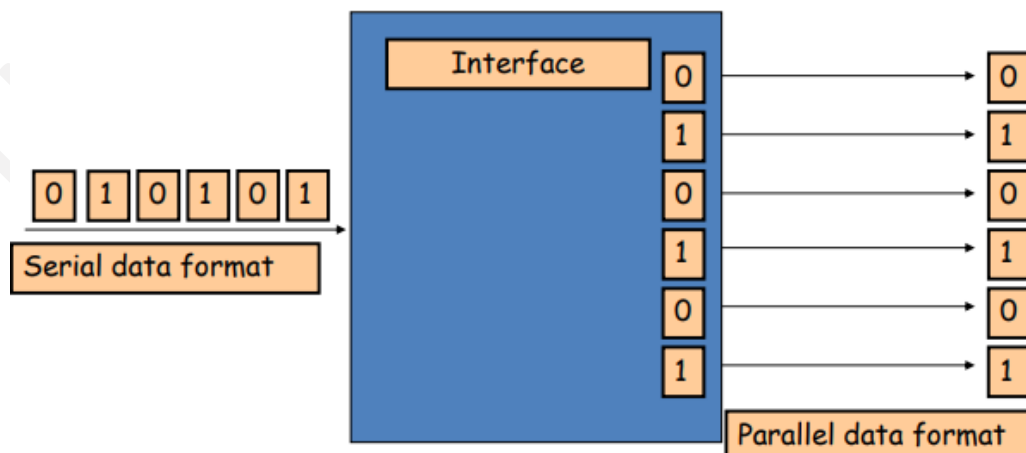
1. A. What is serial and parallel communication? Explain with example.

- **Serial communication** transmits data one bit at a time, sequentially, making it simpler, cheaper, and more suitable for long distances.

Example: Sending data over a serial port (RS-232) connection between a computer and a printer. Each bit of the data (like a character or a command) is sent sequentially, one after the other, over a single wire.

- **Parallel communication** transmits data multiple bits at a time, simultaneously, making it faster, but more complex, expensive, and suitable for short distances.

Example: Sending data between a computer and its memory. In parallel communication, a group of wires is used to transfer multiple bits simultaneously, allowing for faster data transfer compared to serial communication.



Session 2013-2014.

6. B. Describe the process of a device initiated I/O transfer.

Here are the steps of a device-initiated I/O transfer, described briefly:

1. Device Request: The I/O device needs to transfer data.
2. Device Notifies CPU: The device sends an interrupt signal to the CPU.
3. Interrupt Handling: The CPU temporarily stops its current task.
4. Interrupt Service Routine (ISR): The CPU runs a small program to handle the I/O request.
5. I/O Operation Initiation: The CPU sets up the parameters for the data transfer.
6. Data Transfer: Data is transferred between the device and memory.
7. Completion Notification: The device signals the CPU that the transfer is complete.
8. Interrupt Handling (Again): The CPU handles the completion interrupt.
9. Resume Execution: The CPU resumes its previous task

Session 2012-2013.

1. A) What is buffer? Why we use buffers?

A **buffer** is an area of memory used for the transfer of data between a computer and a peripheral.

- A buffer provides temporary storage of data.
- Using a buffer provides a link between a device and the processor and helps compensate for any differences in their working speeds

B) Define interfaces. What are the characteristics of computer peripherals.

The **interface** is the combination of hardware and software needed to link the CPU to the peripherals and to enable them to communicate with the CPU despite all their differing characteristics.

Computer peripherals have different characteristics.

For example, they may:

- Have different data transfer rates;
- Use a wide variety of codes and control signals;
- Transmit data in serial or in parallel form;
- Even work at higher voltages than the CPU; &
- All operate at much slower speeds than the CPU.

C) Differentiate between buffers and spoolers.

- **Buffer v's Spooler:** If the CPU is very busy and doesn't get much idle time then spooling can be a very slow process.
- **Spooler v's Buffer:** A buffer is limited by the amount of RAM whereas a spooler uses backing storage which has a very large capacity.
- Use BOTH for optimum efficiency.

D) What do you mean by solid state storage device? Write down the features and disadvantages of solid state storage devices.

A *solid state storage device* is a type of storage device that uses flash memory to store data. Unlike traditional hard drives, it has no moving parts, which makes it faster and more durable.

Features:

- ✓ Speed: Faster data access and transfer speeds.
- ✓ Durability: More resistant to physical shock and damage.
- ✓ Quiet Operation: No moving parts, so they are silent.
- ✓ Low Power Consumption: Uses less power than traditional hard drives.
- ✓ Compact Size: Smaller and lighter than hard drives.

Disadvantages:

- ✓ Cost: Generally more expensive per gigabyte than traditional hard drives.
- ✓ Limited Write Cycles: Can wear out over time with many write and erase cycles.
- ✓ Capacity: Typically offers less storage capacity compared to the highest-capacity hard drives.

Arpita ma'am.

1. A. What are the different components of IoT? How does the IoT affect our everyday lives?

Components of IoT:

- Devices/Sensors: Gadgets that gather data, like smartwatches and security cameras.
- Connectivity: How these gadgets connect to the internet, using Wi-Fi, Bluetooth, or cellular networks.
- Data Processing: The data is analyzed either on the device, nearby computers, or in the cloud.
- User Interface: How we interact with IoT devices, such as through apps, websites, or voice commands.

How IoT Affects Our Everyday Lives:

- Smart Homes: Control lights, thermostats, and security systems remotely, making life easier and homes more secure.
- Health and Fitness: Wearable devices track activities, heart rate, and sleep to help you stay healthy.
- Transportation: IoT helps with real-time navigation, vehicle monitoring, and even self-driving cars.
- Workplaces: Automate tasks like lighting and heating, making offices more efficient and productive.
- Retail: Improve shopping with smart shelves, personalized ads, and stores without cashiers.
- Agriculture: Farmers use IoT to monitor crops and soil, improving resource use and crop growth.
- Cities: Manage traffic, air quality, and energy use to make city living better.

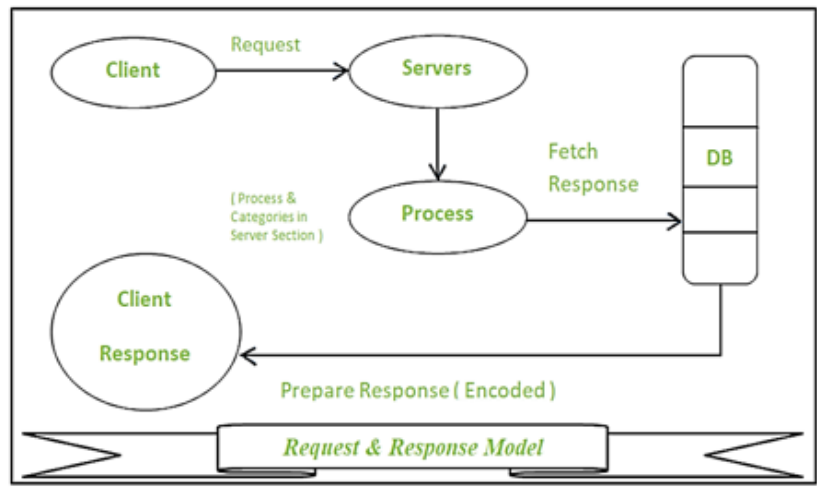
B. What is communication model in IoT? Explain different types of communication model in IoT with diagram and example.

The communication model in IoT explains how IoT devices talk to each other and share information. It includes the different ways and paths that devices use to connect and exchange data.

Different types of communication model:

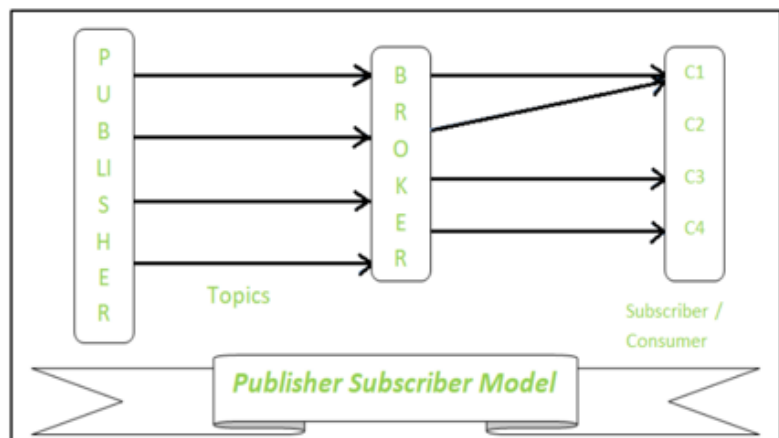
1. Request & Response Model

Example: Sending a spreadsheet to the printer



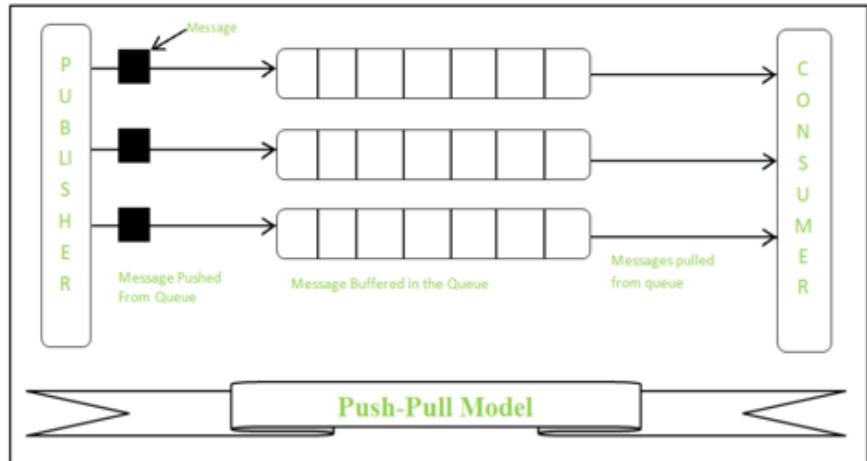
2. Publisher-Subscriber Model

Example: Public sensors with a massive base of uniform users that will use the data.



3. Push-Pull Model

Example: Queues help in decoupling the messaging between the producers and consumers.



4. Exclusive Pair

Example: The WebSocket-based communication API.

